

Bihar Engineering University, Patna**B.Tech. 1st Semester Examination, 2023****Course: B.Tech.****Code: 103102****Subject: Mathematics-I (Calculus & Differential Equations)****Time: 03 Hours****Full Marks: 70****Instructions:-**

- (i) The marks are indicated in the right-hand margin.
 (ii) There are **NINE** questions in this paper.
 (iii) Attempt **FIVE** questions in all.
 (iv) Question No. **1** is compulsory.
 (v) Symbols used (if any) have their usual meanings.

Q.1 Choose the correct answer of the following (Any seven question only):**[2 × 7 = 14]**

- (a) The solution of the given differential equation is :
 $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$
 (i) $x^3 + y^3 - 6xy(x + y) = c$ (ii) $2x^3 + 4y^3 - 4xy = c$
 (iii) $x^3 + y^3 - 5xy(x + y) = c$ (iv) $x^3 + y^3 - 6xy = c$
- (b) The complementary function for the given differential equation is:
 $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = x^2$
 (i) $c_1e^{-x} + c_2e^{4x}$ (ii) $c_1e^{-2x} + c_2e^{-x}$
 (iii) $c_1e^{-3x} + c_2e^{-x}$ (iv) None of the above.
- (c) The solution of the differential equation $\frac{dy}{dx} - y \cot(x) = 2x \sin(x)$ is :
 (i) $y = x^2 + c$ (ii) $y \sin(x) = 2x + c$
 (iii) $y \operatorname{cosec}(x) = x^2 + c$ (iv) $y \cot(x) = 3x + c$
- (d) The wronskian of the functions $\{\sin(x), \cos(x)\}$ is :
 (i) 1 (ii) -1
 (iii) 0 (iv) 2
- (e) Suppose $\vec{A} = x^2z^2\hat{i} - 2y^2z^2\hat{j} + xy^2z\hat{k}$ then the divergence of $\vec{A}$ at point P (1, -1, 1) is :
 (i) 4 (ii) 5
 (iii) 6 (iv) 7
- (f) Let $A = \int_R \sin(x + y) dx dy$, where the region R is given by $\{0 \leq x \leq \frac{\pi}{2}, 0 \leq y \leq \frac{\pi}{2}\}$ then the value of A is :
 (i) -1 (ii) 1
 (iii) 0 (iv) 2
- (g) The series $1 + \frac{3}{2!} + \frac{5}{3!} + \frac{7}{4!} + \dots$ is :
 (i) convergent (ii) divergent
 (iii) Neither convergent nor divergent (iv) None
- (h) If $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} x^2 + \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then
 (i) f is differentiable on $\mathbb{R}$ (ii) f is differentiable on $\mathbb{R} - \{0\}$
 (iii) f is not differentiable at 0 (iv) None of these
- (i) If $f(x, y) = e^{x+y+1}$ then the value of $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$ is
 (i) $(x + y) e^{x+y+1}$ (ii) $xe^y + ye^x$
 (iii) $2e^{x+y+1}$ (iv) None of the above
- (j) The value of $\Gamma\left(-\frac{3}{2}\right)$ is
 (i) 1 (ii) $\frac{\pi}{2}$
 (iii) $\sqrt{\pi}$ (iv) $\frac{1}{3}\sqrt{\pi}$

Q.2 (a) Show that:

$$\int_0^1 (\ln x)^n dx = (-1)^n n!, \quad n \in \mathbb{N}. \quad [7]$$

(b) If α and β are the roots of the equation $ax^2 + bx + c = 0$ evaluate the following limit.

$$\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2} \quad [7]$$

Q.3 (a) Test the convergence of

$$\frac{1}{2\sqrt{1}} + \frac{1}{3\sqrt{2}}x^2 + \frac{1}{4\sqrt{3}}x^4 + \frac{1}{5\sqrt{4}}x^6 + \dots \quad [8]$$

(b) Prove that

$$\text{Curl}(fv) = (\nabla f) \times v + f(\text{curl } v)$$

Q.4 (a) Use Taylor's theorem to prove that

$$1 + \frac{x}{2} - \frac{x^3}{8} < \sqrt{1+x} < 1 + \frac{x}{2}, \text{ if } x > 0. \quad [8]$$

(b) If $p(x)$ is a polynomial of degree > 1 and $k \in \mathbb{R}$, prove that between any two real roots of $p(x) = 0$ there exists a real root of $p'(x) + kp(x) = 0$. [6]

Q.5 (a) Show that the function $f(x, y) = \begin{cases} \frac{x^3 + y^3}{x - y}, & x \neq y \\ 0, & x = y \end{cases}$ is not continuous at origin. But f_x and f_y exists at $(0, 0)$. [6]

(b) Show that $\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{\Gamma(\frac{m+1}{2})\Gamma(\frac{n+1}{2})}{2\Gamma(\frac{m+n+2}{2})}$. [8]

Q.6 (a) Find the maxima and minima of the function $u = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. [7]

(b) Find $u(x, y) = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, $0 < x, y < 1$, then prove that [7]

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0.$$

Q.7 (a) Find the Fourier series to represent the function defined as [6]

$$f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi < x < 0 \\ \frac{\pi}{2} - x, & 0 < x < \pi \end{cases}$$

(b) Solve the following differential equations, [4x2=8]

(i) $(x+2)^2 \frac{d^2y}{dx^2} - (x+2) \frac{dy}{dx} - 3y = 0$

(ii) $(y - 1)dx + x(x + 1)dy = 0$

Q.8 (a) Use variation of parameters method to solve [7]

$$(D^3 + 1)y = e^{x/2} \sin \frac{\sqrt{3}}{2}x$$

(b) Using operator method solve the following differential equation, [7]

$$(D^2 - 2D + 5)y = e^x \sin 2x$$

Q.9 (a) Find a complete integral of the following equation [7]

$$p^2x + q^2y = z$$

(b) Find the general solution of the partial differential equation [7]

$$x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = (x + y)z$$